

Energy efficiency in farming means upgrading equipment and optimising processes to use less energy to grow crops and raise livestock. Benefits can include:

- reduced energy costs
- protection against unexpected rises in energy prices
- improved energy security
- lower emissions from farm activities.

What you can do

To improve your energy efficiency, you may want to explore:

- understanding your energy use
- changes to practices to reduce fuel and energy use
- upgrading parts of existing machinery
- upgrading machinery and installing energy efficient technology
- on farm energy generation.

Understand your energy use

Becoming more energy efficient is a chance to help reduce emissions in farming and land management. While it may be a smaller part of agricultural emissions overall, in some sub-sectors—like poultry farming, irrigated crops, horticulture, and land-based aquaculture—energy use can be a large source of emissions.

Most emissions from energy use on farm comes from the combustion of diesel through using tractors, light vehicles, water pumps, and heating and cooling. This contributes to a farm's overall emissions while also becoming one of the agriculture sector's highest input costs.

The first step to becoming more energy efficient is looking at how you currently use energy. See the Queensland Government's [energy savings ideas for agricultural businesses](#) for practical information on understanding your energy use.

Changes to practices to reduce fuel and energy use

Certain farming techniques can improve energy efficiency by reducing fuel use. For example, you can:

- implement efficient vehicle operation practices, like efficient driving, tractor set up and maintenance
- optimise field operations through planning or the use of precision agriculture equipment. Precision agriculture uses technology to find variations in different areas of land and use this information to tailor management responses
- introduce or expand minimum tillage practises
- retain stubble after harvest.

Learn more about [how to save on fuel costs](#) (Queensland Ag Energy Hub) and [precision agriculture](#) (AgriFutures).

Upgrade parts of existing machinery

Most farm machinery uses diesel as fuel. It produces greenhouse gas emissions when combusted. There are different ways you can replace or upgrade existing parts of machinery to use less diesel. Examples include:

- swapping to lower rolling resistance tyres, to reduce energy loss
- upgrading engine components to improve fuel efficiency
- installing GPS automation in tractors and machinery to control steering, spraying, and harvesting.

See more information on upgrading parts of machinery to [reduce energy use and costs](#) (NSW Government).

Upgrade machinery and install energy efficient technology

A lot of energy is used to heat, cool, and provide water to infrastructure and farming equipment. Upgrading machinery and installing energy efficient technology can help reduce energy consumption.

Ways to improve energy efficiency in infrastructure and farming equipment include:

- improving irrigation and pumping to minimise water losses, such as using infield sensors to match water use with water requirements for your crops
- assessing pump set up and maximising pump efficiency to reduce fuel consumption
- increasing building insulation to reduce the amount of energy used to heat or cool buildings
- using precision sprayers, to apply fertilisers, pesticides and herbicides only when and where needed
- swapping diesel-powered light vehicles for electric vehicles.

This will help improve the energy efficiency of your farming operations.

While upgrading existing machinery requires some investment, you can see returns in reduced input and running costs.

See more information on [electric or farm efficient vehicles](#) (CEFC).

A real example: optical spot spraying

You can read about optical spot spraying, an example of energy efficient technology, on the [GRDC](#) website.

On-farm energy generation

Generating your own energy on farm, such as through solar panels and small-scale wind turbines, can reduce your reliance on the wider electricity networks, reducing your energy bill. It can also diversify income streams when selling excess energy generated back into the wider grid network.

There are also opportunities to install energy efficient infrastructure and equipment that can be powered by renewable energy, for example:

- upgrading hot water systems to solar powered alternatives (such as solar thermal systems or energy efficient heat pumps)
- using solar pumps for small-scale water use, such as providing water for livestock
- using energy efficient infrastructure to provide heat and cool requirements for livestock industries (such as glycol chillers milking sheds).

See EvokeAG's [renewable energy in agriculture](#) for information on the benefits, challenges and options for renewable energy on your farm.

A real example: Avondale Feedlot

Learn about how Avondale Feedlot are using an [electrified off-grid system](#) to power their enterprise (NSW Government).

Resources to help you improve your energy efficiency

Explore further key resources on improving your energy efficiency.

State and territory resources:

Some state governments have resources to help you assess your current energy use and find activities to improve energy efficiency:

- New South Wales: [On-farm energy](#)
- Queensland: [Energy savings ideas for agricultural businesses](#)
- Tasmania: [Improving energy efficiencies on-farm](#)
- Victoria: [Energy use on farms](#)

See more

The CEFC and National Farmers Federation have produced [a practical guide to lowering on-farm energy use and carbon emissions](#).

Learn about energy efficiency:

- [CEFC](#): Find out more about activities to improve energy efficiency.
- [DCCEEW](#): Find commodity specific information, programs and case studies.

- **Energy Smart Farming:** Find resources, case studies and grants.

Source: <https://www.agriculture.gov.au/agriculture-land/farm-food-drought/climatechange/carbon-farming-outreach-program/hub/ways-reduce-emissions/improve-energy-efficiency>